

DePIN

Decentralised physical infrastructure

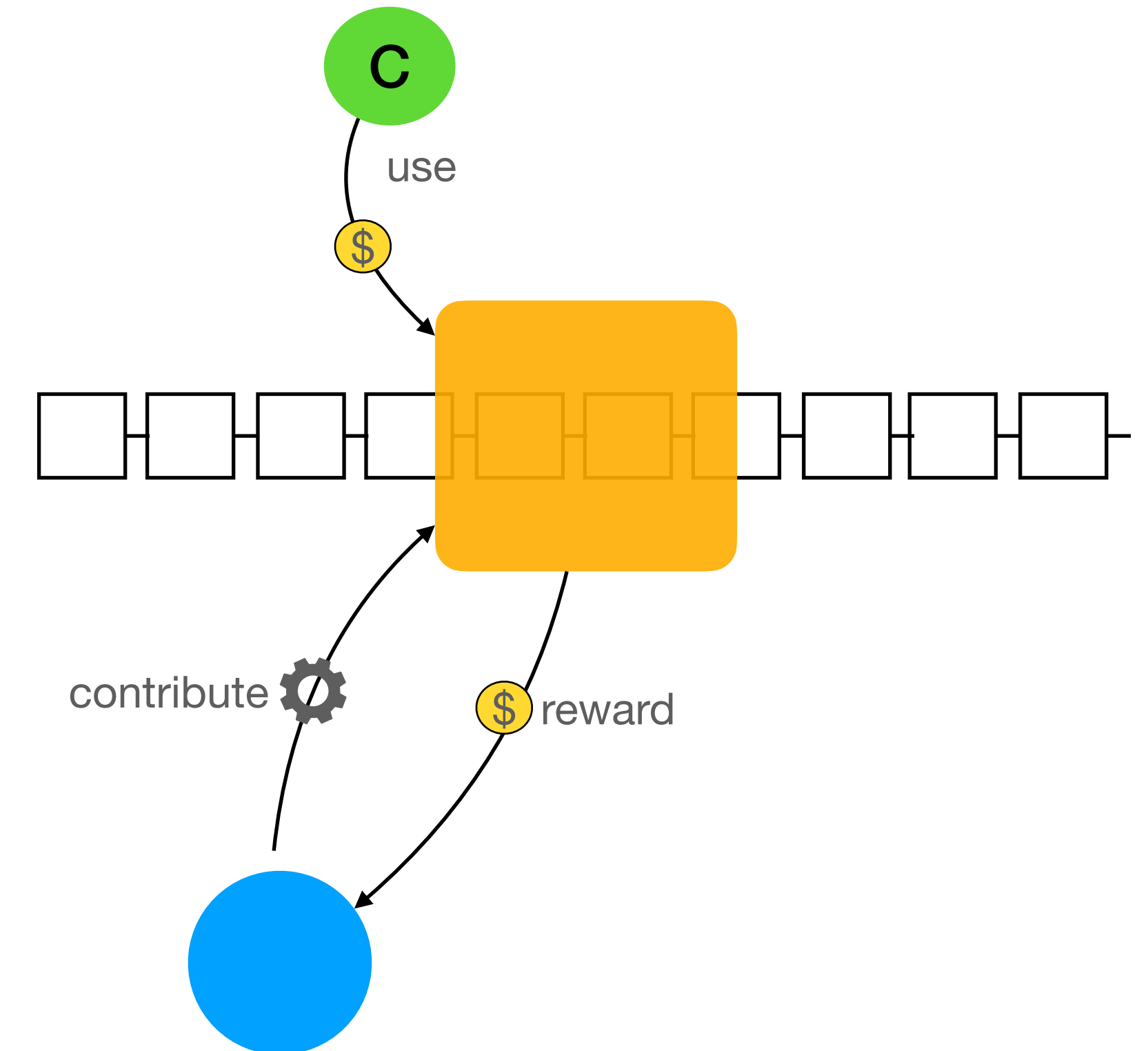
Leander Jehl

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Decentralized physical infrastructure

Idea:

- Different people contribute infrastructure to a service,
storage space, sensor data, network bandwidth, GPU resources
- Contributors receive rewards in tokens.
- Users pay tokens for use of infrastructure.



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Decentralized physical infrastructure

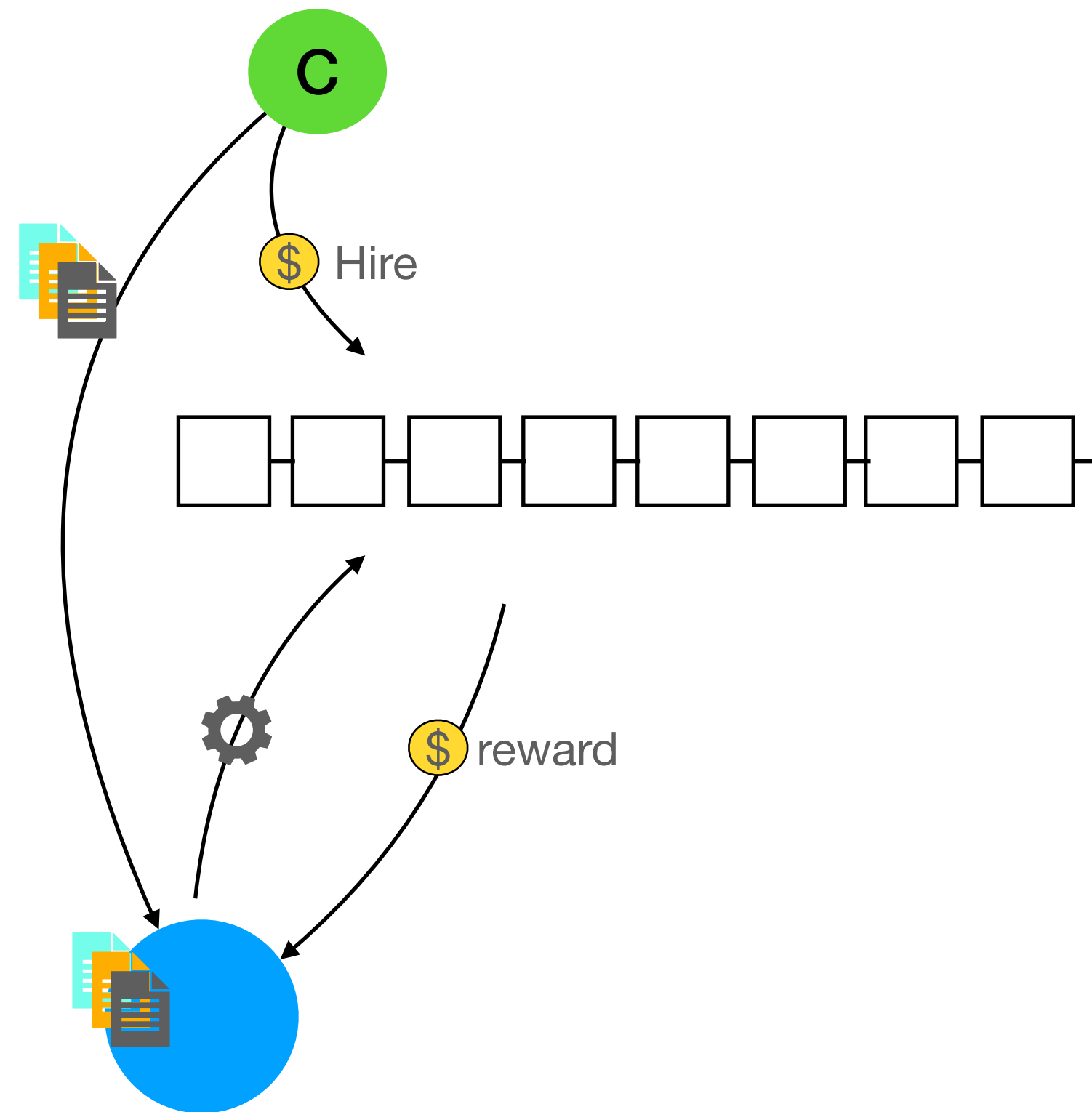
Example:

- Run a storage server on FileCoin, Storej, ...
- Drive around with a dashcam for google maps like data
- Sell your browser history for advertisement
- See ads in your browser for rewards

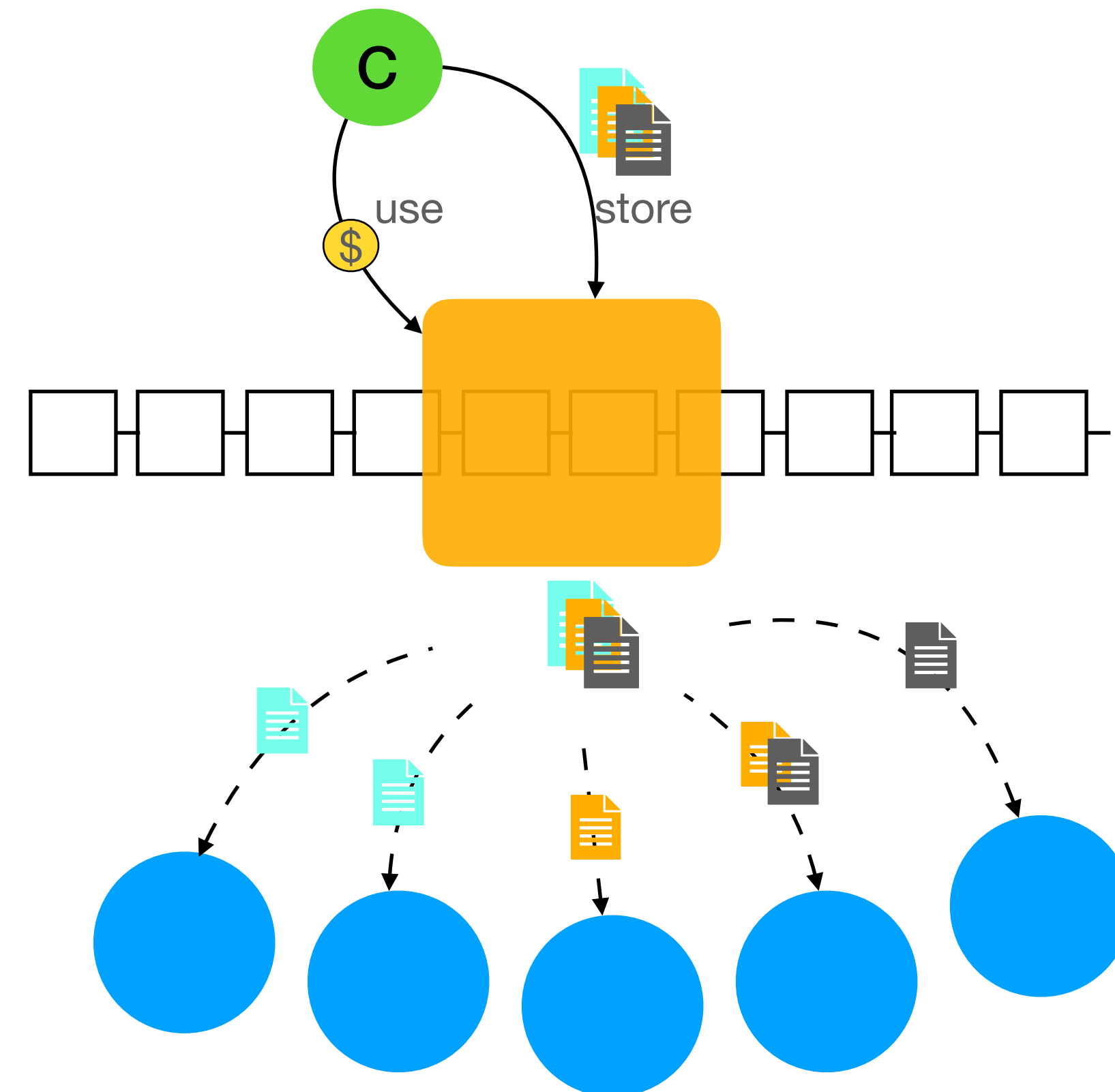
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Decentralized physical infrastructure

Hire individual contributors



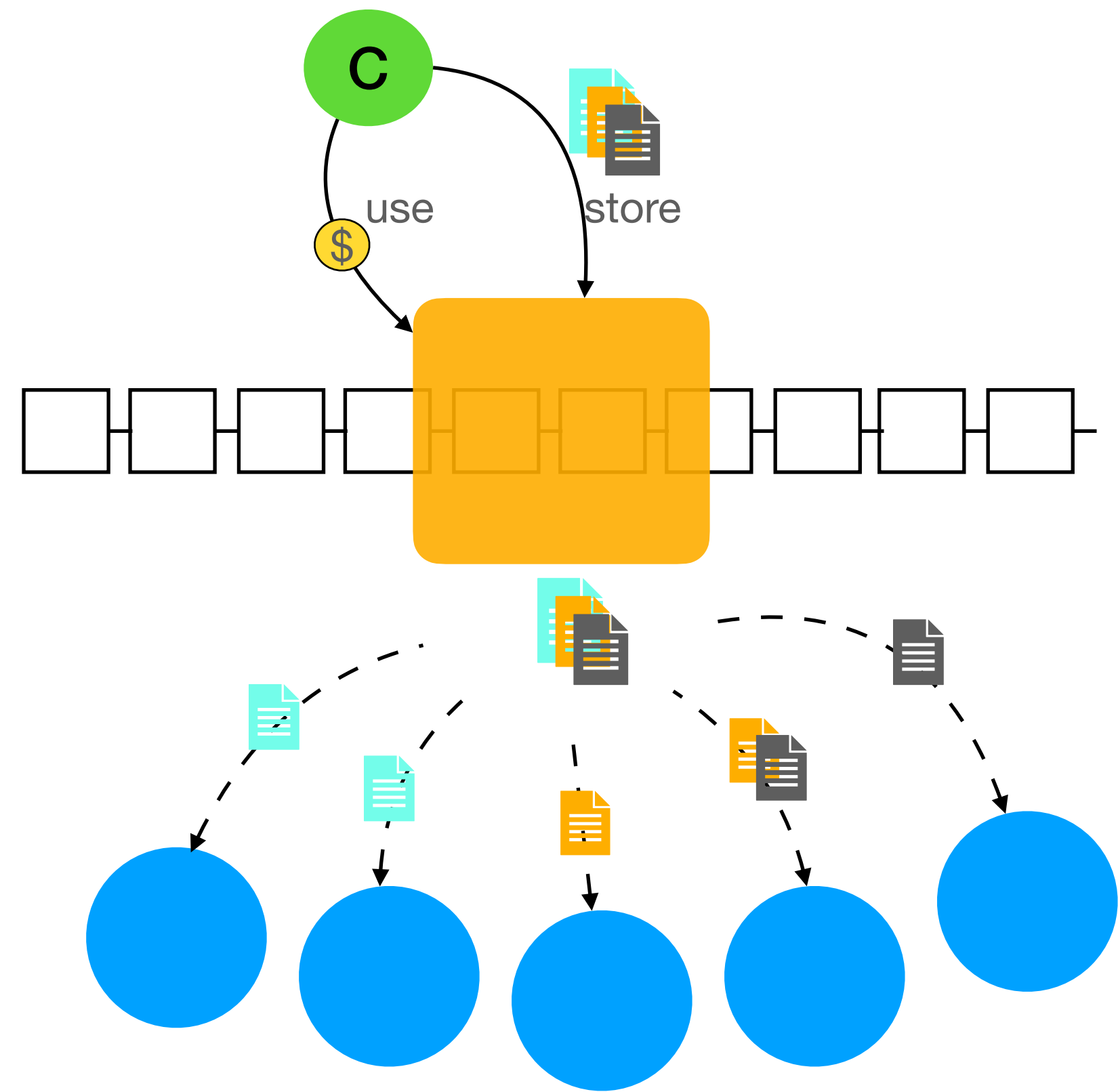
Pay the system



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Problems

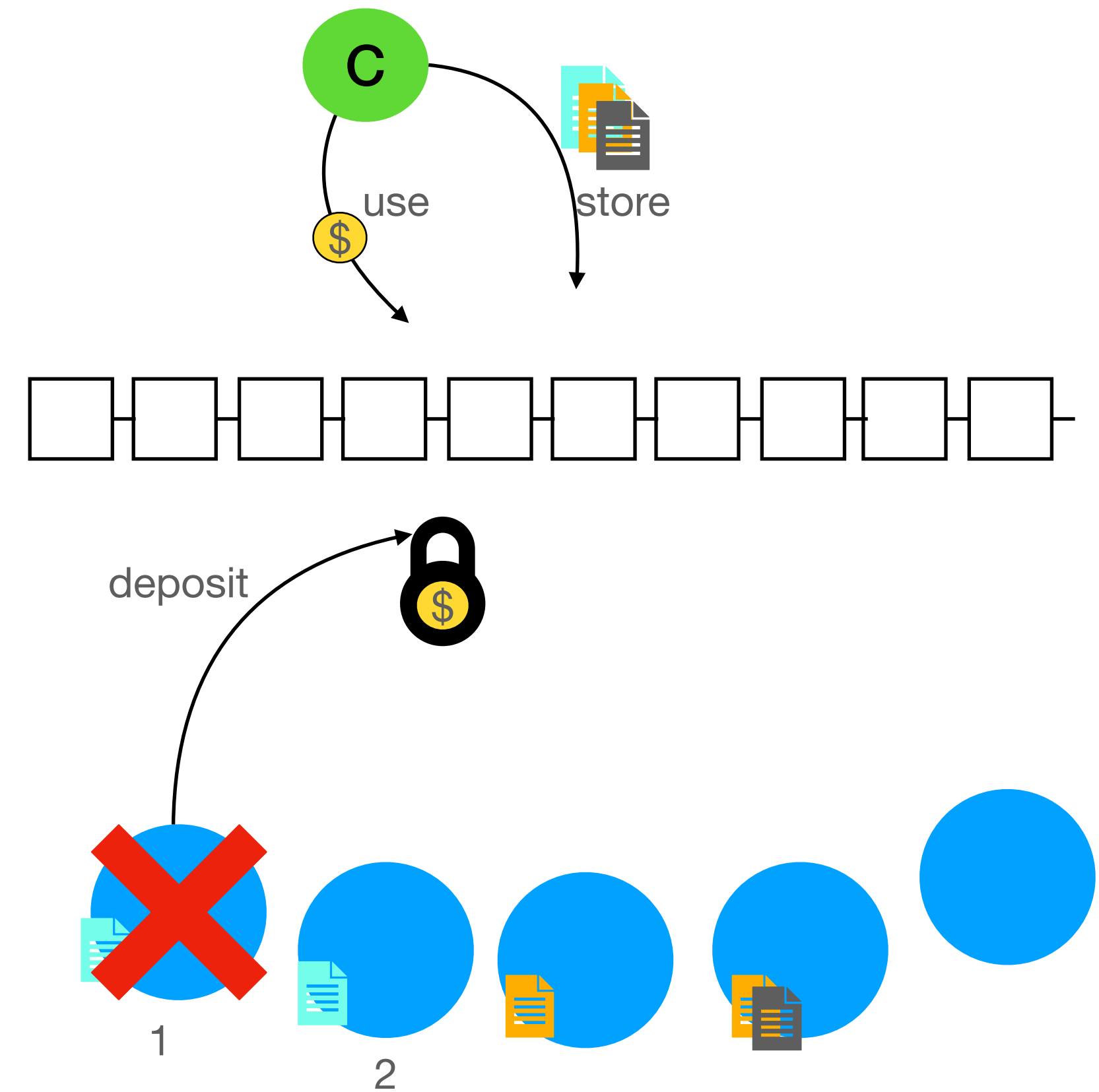
- Contributors leave
- Contributors do not contribute correctly
- Varying customers
- Speculation



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Problems

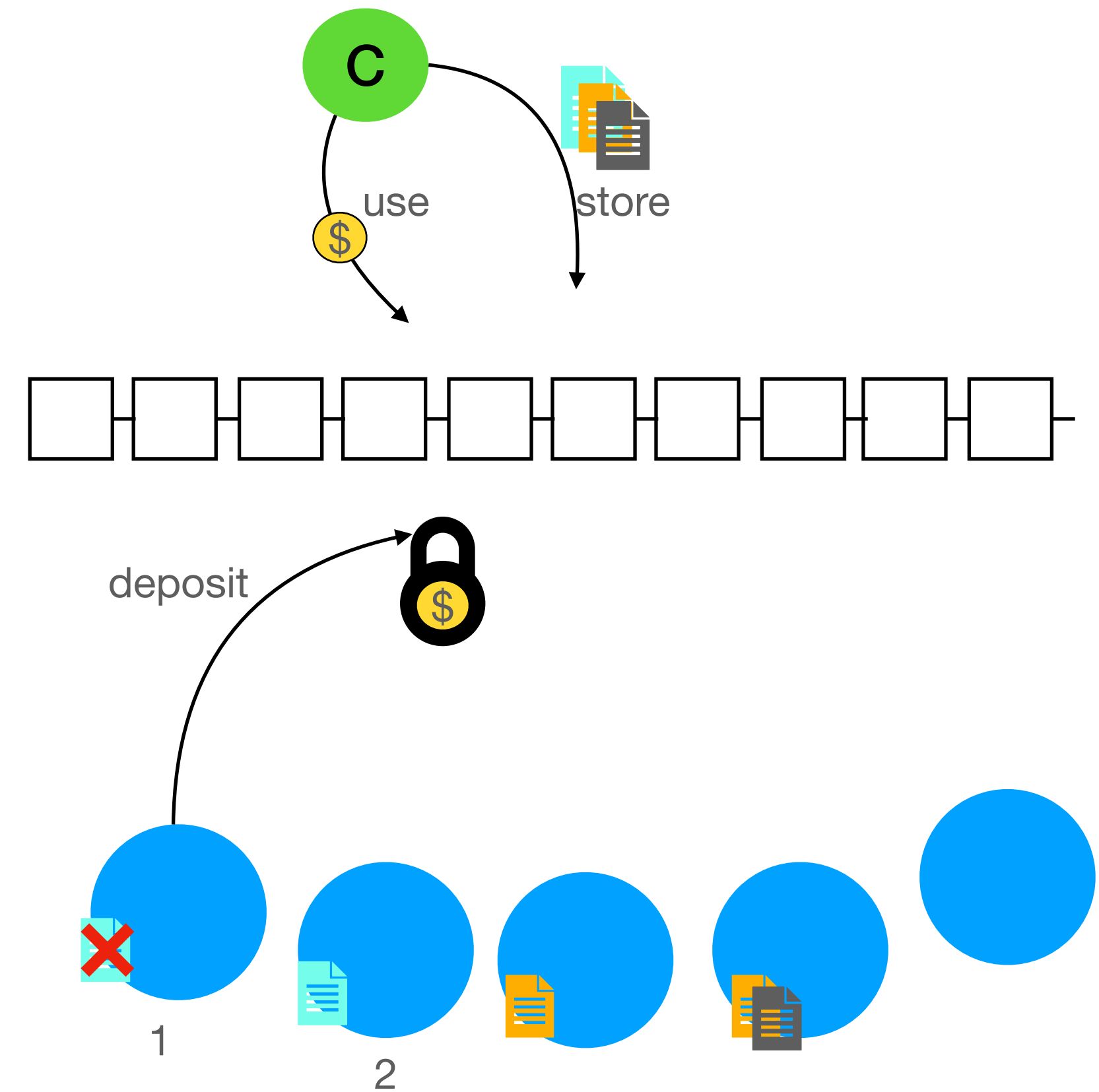
- Contributors leave
 - redundancy
 - contributors deposit stake



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Problems

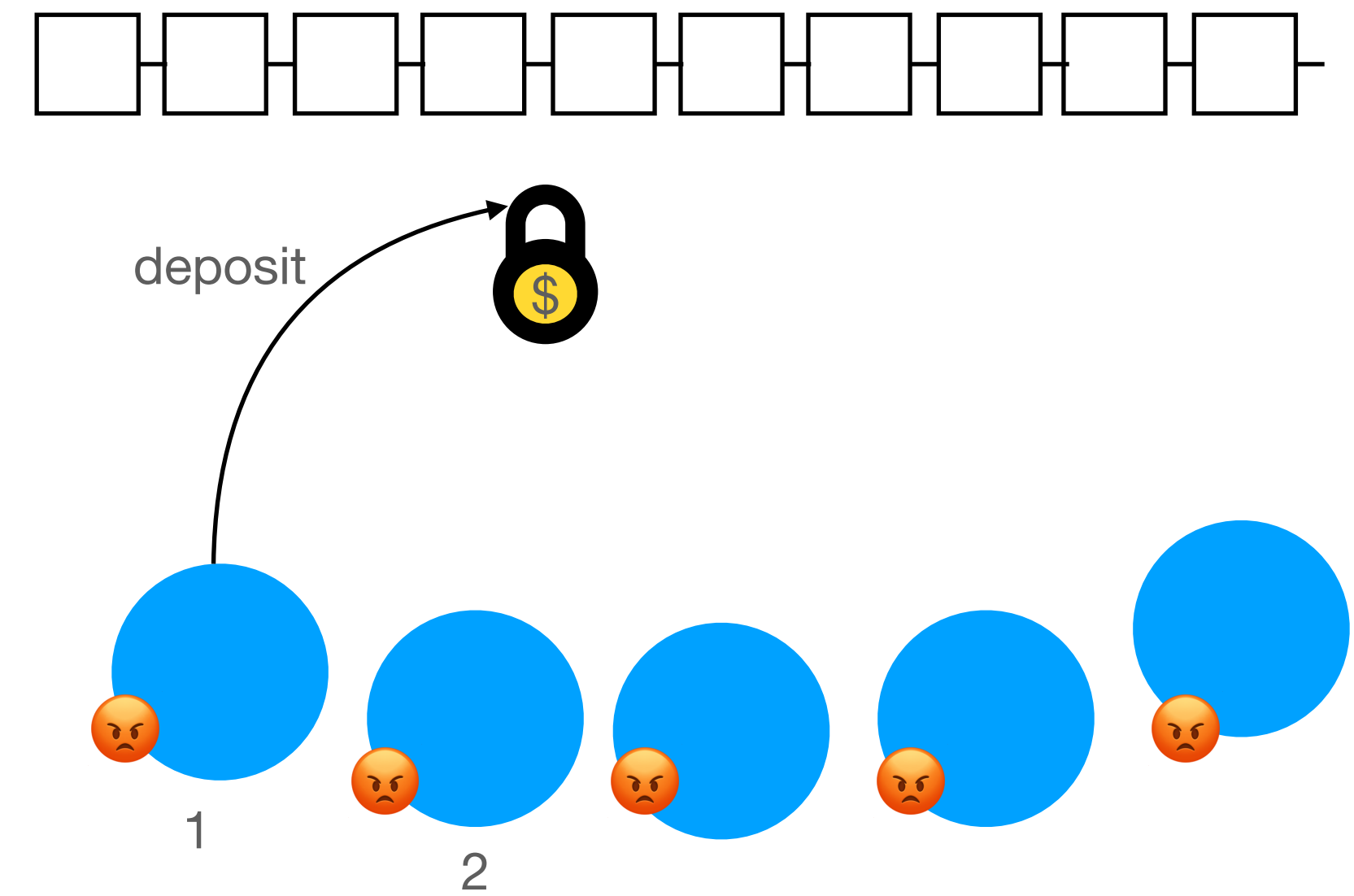
- do not contribute correctly
 - redundancy
- contributors deposit stake
- need to be able to check contribution



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Problems

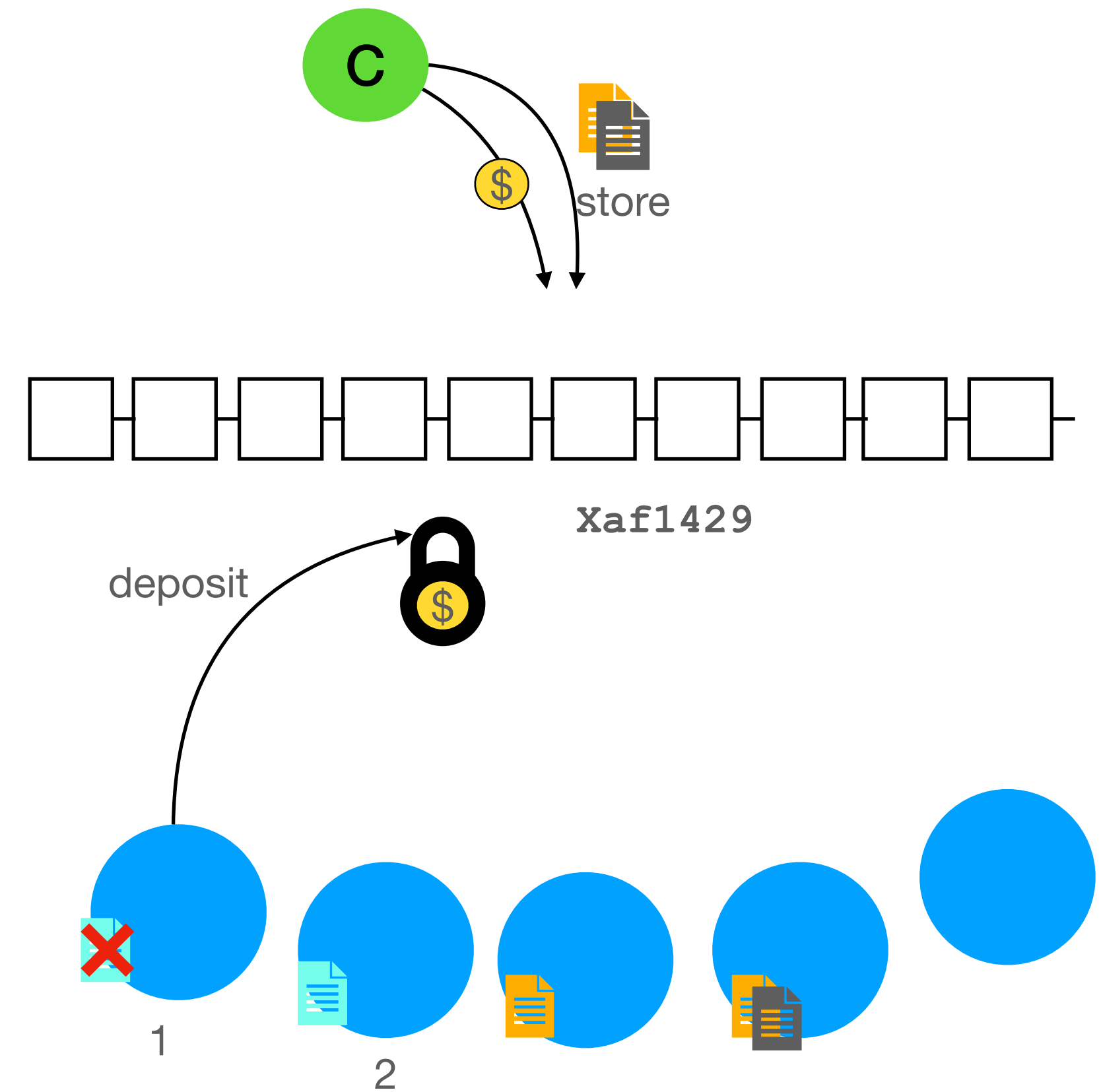
- varying customers
 - mint coins for reward



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Problems

- Checking contribution
 - Example:
content addressed storage
hash addressed storage
- Regular checks by
 - Client or Smart contract



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Decentralized physical infrastructure

Summary

- Storage, compute resources similar to cloud computing
- But need
 - Redundancy
 - Checks
 - Many transactions (fees)
- Expansive and less reliable

Use case

- Blockchain related data or computation
- Cheaper than on chain, similar failure model similar censorship resistance